Physics 110B Winter 2018 Midterm

Name:

Perm:

BASIC EQUATIONS OF ELECTRODYNAMICS

Maxwell's Equations

In General:

$$\vec{\nabla} \cdot \vec{E} = \frac{1}{\varepsilon_0}\rho$$

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\vec{\nabla} \cdot \vec{B} = 0$$

$$\vec{\nabla} \times \vec{B} = \mu_0 \vec{J} + \mu_0 \varepsilon_0 \frac{\partial \vec{E}}{\partial t}$$

In Matter:

$$\vec{\nabla} \cdot \vec{D} = \rho_f$$

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\vec{\nabla} \cdot \vec{B} = 0$$

$$\vec{\nabla} \times \vec{H} = \vec{J}_f + \frac{\partial \vec{D}}{\partial t}$$

Auxiliary Fields

Definitions:

$$\vec{D} = \varepsilon_0 \vec{E} + \vec{P}$$

$$\vec{H} = \frac{1}{\mu_0}\vec{B} - \vec{M}$$

Linear Media:

$$\vec{P} = \varepsilon_0 x_e \vec{E}, \quad \vec{D} = \varepsilon \vec{E}$$

$$\vec{M} = x_m \vec{H}, \quad \vec{H} = \frac{1}{\mu}\vec{B}$$

Potentials

$$\vec{E} = -\vec{\nabla} V - \frac{\partial \vec{A}}{\partial t}, \quad \vec{B} = \vec{\nabla} \times \vec{A}$$

Lorentz Force Law

$$\vec{F} = q\left(\vec{E} + \vec{v} \times \vec{B}\right)$$

Energy

Energy: $U = \frac{1}{2}\int\left(\varepsilon_0 E^2 + \frac{1}{\mu_0} B^2\right)d\tau$

FUNDAMENTAL CONSTANTS

$\varepsilon_0 = 8.85 \times 10^{-12}\, C^2 / Nm^2$

(permittivity of free space)

$\mu_0 = 4\pi \times 10^{-7}\, \mathrm{N/A^2}$

(permeability of free space)

$e = 1.60 \times 10^{-19}\, C$

(charge of the electron)

$m = 9.11 \times 10^{-31}\, kg$

(mass of the electron)

SPHERICAL AND CYLINDRICAL COORDINATES

Spherical

$$x = r\sin\theta\cos\phi \qquad \hat{x} = \sin\theta\cos\phi\,\hat{r} + \cos\theta\cos\phi\,\hat{\theta} - \sin\phi\,\hat{\phi}$$

$$y = r\sin\theta\sin\phi \qquad \hat{y} = \sin\theta\sin\phi\,\hat{r} + \cos\theta\sin\phi\,\hat{\theta} + \cos\phi\,\hat{\phi}$$

$$z = r\cos\theta \qquad \hat{z} = \cos\theta\,\hat{r} - \sin\theta\,\hat{\theta}$$

Cylindrical

$$x = s\cos\phi \qquad \hat{x} = \cos\phi\,\hat{s} - \sin\phi\,\hat{\phi}$$

$$y = s\sin\phi \qquad \hat{y} = \sin\phi\,\hat{s} + \cos\phi\,\hat{\phi}$$

$$z = z \qquad \hat{z} = \hat{z}$$

FUNDAMENTAL THEOREMS

Gradient Theorem: $\int_a^b (\nabla f) \cdot d\vec{l} = f\left(\vec{b}\right) - f(\vec{a})$

Divergence Theorem: $\int \left(\nabla \cdot \vec{A}\right) d\tau = \oint \vec{A} \cdot d\vec{a}$

Curl Theorem: $\int \left(\vec{\nabla} \times \vec{A}\right) \cdot d\vec{a} = \oint \vec{A} \cdot d\vec{l}$

VECTOR DERIVATIVES

Cartesian. $d\vec{l} = dx\,\hat{x} + dy\,\hat{y} + dz\,\hat{z}$; $\;d\tau = dx\,dy\,dz$

Gradient: $\vec{\nabla} t = \frac{\partial t}{\partial x}\hat{x} + \frac{\partial t}{\partial y}\hat{y} + \frac{\partial t}{\partial z}\hat{z}$

Divergence: $\vec{\nabla} \cdot v = \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z}$

Curl: $\vec{\nabla} \times v = \left(\frac{\partial v_z}{\partial y} - \frac{\partial v_y}{\partial z}\right)\hat{x} + \left(\frac{\partial v_x}{\partial z} - \frac{\partial v_z}{\partial x}\right)\hat{y} + \left(\frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y}\right)\hat{z}$

Laplacian: $\nabla^2 t = \frac{\partial^2 t}{\partial x^2} + \frac{\partial^2 t}{\partial y^2} + \frac{\partial^2 t}{\partial z^2}$

Spherical. $d\vec{l} = dr\,\hat{r} + r\,d\theta\,\hat{\theta} + r\sin\theta\,d\phi\,\hat{\varphi}$;

$d\tau = r^2 \sin\theta\; dr\; d\theta\; d\phi$

Gradient: $\vec{\nabla} t = \frac{\partial t}{\partial r}\hat{r} + \frac{1}{r}\frac{\partial t}{\partial \theta}\hat{\theta} + \frac{1}{r\sin\theta}\frac{\partial t}{\partial \phi}\hat{\phi}$

Divergence:

$$\vec{\nabla} \cdot v = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 v_r\right) + \frac{1}{r\sin\theta}\frac{\partial}{\partial \theta}\left(\sin\theta\, v_\theta\right) + \frac{1}{r\sin\theta}\frac{\partial v_\phi}{\partial \phi}$$

Curl:

$$\vec{\nabla} \times v = \frac{1}{r\sin\theta}\left[\frac{\partial}{\partial\theta}\left(\sin\theta\, v_\varphi\right) - \frac{\partial v_\theta}{\partial\varphi}\right]\hat{r} + \frac{1}{r}\left[\frac{1}{\sin\theta}\frac{\partial v_r}{\partial\varphi} - \frac{\partial}{\partial r}\left(r\, v_\varphi\right)\right]\hat{\theta} + \frac{1}{r}\left[\frac{\partial}{\partial r}\left(r\, v_\theta\right) - \frac{\partial v_r}{\partial\theta}\right]\hat{\varphi}$$

Laplacian: $\nabla^2 t = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial t}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial t}{\partial\theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2 t}{\partial\varphi^2}$

Cylindrical. $d\vec{l} = ds\,\hat{s} + s\,d\phi\,\hat{\phi} + dz\,\hat{z}$; $\;d\tau = s\,ds\,d\phi\,dz$

Gradient: $\vec{\nabla} t = (\partial t/\partial s)\,\hat{s} \;+\; (1/s)(\partial t/\partial\varphi)\,\hat{\varphi} \;+\; (\partial t/\partial z)\,\hat{z}$

Divergence: $\vec{\nabla} \cdot v = \frac{1}{s}\frac{\partial}{\partial s}\left(s v_s\right) + \frac{1}{s}\frac{\partial v_\phi}{\partial \phi} + \frac{\partial v_z}{\partial z}$

Curl: $\vec{\nabla} \times v = \left[\frac{1}{s}\frac{\partial v_z}{\partial\phi} - \frac{\partial v_\phi}{\partial z}\right]\hat{s} + \left[\frac{\partial v_s}{\partial z} - \frac{\partial v_z}{\partial s}\right]\hat{\phi} + \frac{1}{s}\left[\frac{\partial}{\partial s}\left(s\, v_\phi\right) - \frac{\partial v_s}{\partial\phi}\right]\hat{z}$

Laplacian: $\nabla^2 t = \frac{1}{s}\frac{\partial}{\partial s}\left(s\frac{\partial t}{\partial s}\right) + \frac{1}{s^2}\frac{\partial^2 t}{\partial\phi^2} + \frac{\partial^2 t}{\partial z^2}$

VECTOR IDENTITIES

Triple Products

(1) $\vec{A} \cdot \left(\vec{B} \times \vec{C}\right) = \vec{B} \cdot \left(\vec{C} \times \vec{A}\right) = \vec{C} \cdot \left(\vec{A} \times \vec{B}\right)$

(2) $\vec{A} \times \left(\vec{B} \times \vec{C}\right) = \vec{B}\left(\vec{A} \cdot \vec{C}\right) - \vec{C}\left(\vec{A} \cdot \vec{B}\right)$

Product Rules

(3) $\vec{\nabla}\left(fg\right) = f\left(\vec{\nabla} g\right) + g\left(\vec{\nabla} f\right)$

(4) $\vec{\nabla}\left(\vec{A} \cdot \vec{B}\right) = \vec{A} \times \left(\vec{\nabla} \times \vec{B}\right) + \vec{B} \times \left(\vec{\nabla} \times \vec{A}\right) + \left(\vec{A} \cdot \vec{\nabla}\right)\vec{B} + \left(\vec{B} \cdot \vec{\nabla}\right)\vec{A}$

(5) $\vec{\nabla} \cdot \left(f\vec{A}\right) = f\left(\vec{\nabla} \cdot \vec{A}\right) + \vec{A} \cdot \left(\vec{\nabla} f\right)$

(6) $\vec{\nabla} \cdot \left(\vec{A} \times \vec{B}\right) = \vec{B} \cdot \left(\vec{\nabla} \times \vec{A}\right) - \vec{A} \cdot \left(\vec{\nabla} \times \vec{B}\right)$

(7) $\vec{\nabla} \times \left(f\vec{A}\right) = f\left(\vec{\nabla} \times \vec{A}\right) - \vec{A} \times \left(\vec{\nabla} f\right)$

(8) $\vec{\nabla} \times \left(\vec{A} \times \vec{B}\right) = \left(\vec{B} \cdot \vec{\nabla}\right)\vec{A} - \left(\vec{A} \cdot \vec{\nabla}\right)\vec{B} + \vec{A}\left(\vec{\nabla} \cdot \vec{B}\right) - \vec{B}\left(\vec{\nabla} \cdot \vec{A}\right)$

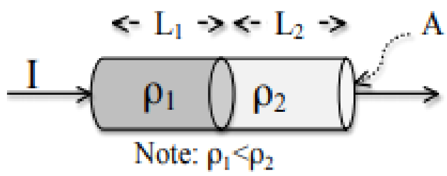
Second Derivatives

(9) $\vec{\nabla} \cdot \left(\vec{\nabla} \times \vec{A}\right) = 0$

(10) $\vec{\nabla} \times (\vec{\nabla} f) = 0$

(11) $\vec{\nabla} \times \left(\vec{\nabla} \times \vec{A}\right) = \vec{\nabla}(\vec{\nabla} \cdot \vec{A}) - \nabla^2 \vec{A}$

Problem 1 A cylindrical resistor has a steady current I flowing uniformly through it. The material is uniform in cross section (area A), but the left portion (length L_1, uniform resistivity ρ_1) is made of a different material than the right portion (length L_2, and higher resistivity ρ_2).



(a) (10 points) **What** is the total power dissipated in this resistor in terms of the given quantities?

(b) (5 points) **Where, if anywhere**, inside the conducting region of this resistor, is $\nabla \cdot \vec{J} \neq 0$? If you think $\nabla \cdot \vec{J} \neq 0$ somewhere, what is its sign (positive or negative)?

(c) (5 points) **Where, if anywhere**, inside the conducting region of this resistor, is $\nabla \cdot \vec{E} \neq 0$? If you think $\nabla \cdot \vec{E} \neq 0$ somewhere, what is its sign (positive or negative)?

Problem 2 An infinitely long solenoid along the z-axis has n turns per unit length. If the current in the solenoid is I_1, the magnitude of the magnetic field inside the solenoid is given by $B = \mu_0 n I_1$. A small loop of wire is placed entirely inside the solenoid, and it lies within the xy-plane. The small loop has a cross sectional area A, resistance R, and self-inductance L.

(a) (5 points) When the current in the solenoid is I_1, **find** the absolute value of the magnetic flux through the small loop of wire.

(b) (5 points) Suppose the current in the solenoid is zero, and the current in the small loop of wire is I_2. **Find** the absolute value of the total magnetic flux through the solenoid.

(c) (10 points) Let the current in the solenoid $I_1(t)$ vary slowly with time. **Write** a differential equation relating the current in the small loop $I_2(t)$ to the current in the solenoid $I_1(t)$ in terms of the given quantities. Assuming $I_1(t) = kt$ where k is constant, **solve** for $I_2(t)$ for arbitrary initial conditions.

(d) (5 points) Assume the current in the solenoid is $I_1(t) = kt$. Consider times long after the current has started flowing ($t \gg 0$). **What** is the absolute value of I_2? **What** will be the induced EMF in the solenoid due to the current I_2?

Problem 3 Consider two perfectly conducting cylinders centered on the z-axis: an inner conducting cylinder of radius $s = a$ and an outer conducting cylinder of radius $s = b$ with $b > a$. In cylindrical polar coordinates, the radial unit vector is $\hat{\mathbf{s}}$ and the azimuthal unit vector is $\hat{\phi}$. The region between the cylinders $a < s < b$ is vacuum and the electric and magnetic fields are given by

$$\mathbf{E} = \frac{\cos(kz - \omega t)}{s}\hat{\mathbf{s}} \tag{1}$$

$$\mathbf{B} = \frac{\cos(kz - \omega t)}{cs}\hat{\phi} \tag{2}$$

(a) (5 points) The electric and magnetic fields given by (1) and (2) describe a propagating electromagnetic wave. **What** is the direction of propagation?

(b) (10 points) **Calculate** the *time-averaged* Poynting vector $\langle \vec{S} \rangle = \frac{1}{\mu_0}\langle \vec{E} \times \vec{B} \rangle$ as a function of s for the region $a < s < b$. (*Hint*: find the Poynting vector and average it over one oscillation cycle.)

(c) (5 points) **Calculate** the total angular momentum $\vec{L}$ (with respect to the origin) of the electromagnetic field for the region $-z_0 < z < z_0$?

Problem 4 A left-handed circularly polarized plane wave of light has an electric field given by

$$\mathbf{E}_L = \cos(k_L z - \omega t)\hat{\mathbf{x}} - \sin(k_L z - \omega t)\hat{\mathbf{y}} \tag{3}$$

and a right-handed circularly polarized plane wave of light has an electric field given by

$$\mathbf{E}_R = \cos(k_R z - \omega t)\hat{\mathbf{x}} + \sin(k_R z - \omega t)\hat{\mathbf{y}} \tag{4}$$

(a) (5 points) **What** is the direction of propagation for the light given in (3) and (4)?

(b) (10 points) **Find** the corresponding magnetic fields for the electric fields given by (3) and (4).

(c) (5 points) In a plasma in a magnetic field, left and right-handed circularly polarized light can travel at different speeds. Assume we fill all of space with such a plasma in a magnetic field, and assume the medium is linear. At the frequency ω, let the phase velocity of circularly polarized light for the material be given by v_L and v_R for the left and right-handed cases, respectively. **Express** the wavenumbers k_L and k_R in terms of v_L, v_R, and ω.

(d) (5 points) Suppose we have a *linearly* polarized plane wave of light traveling in the $+\hat{\mathbf{z}}$ direction. In the xy-plane ($z = 0$) its electric field is

$$\mathbf{E} = E_0 \cos(-\omega t)\hat{\mathbf{x}} \tag{5}$$

Express this electric field of *linearly* polarized light as a superposition of the electric fields of left and right-handed polarized light $\vec{E}_L$ and $\vec{E}_R$ given by (3) and (4) for $z \geq 0$.

(e) (10 points) After propagating a short distance d from $z = 0$ to $z = d$ where $d > 0$, the *linearly* polarized plane wave remains *linearly* polarized, but its polarization vector is rotated by an angle θ. **Find** the absolute value of θ in terms of d, k_L, and k_R. (*Hint*: Some useful identities are $\cos\alpha + \cos\beta = 2\cos\left(\frac{\alpha+\beta}{2}\right)\cos\left(\frac{\alpha-\beta}{2}\right)$ and $\sin\alpha - \sin\beta = 2\sin\left(\frac{\alpha-\beta}{2}\right)\cos\left(\frac{\alpha+\beta}{2}\right)$).

Additional space

Additional space

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